Cell phone Price Prediction Analysis Project Report

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Date: 30.03.2026

1. Project Overview

This project is to analyse the relationship between mobile features and their price range, and derive meaningful insights for decision-making

2. Dataset Summary

Rows : 2000
Coloumns : 21
No missing data

3. Tools & Technologies

Python (Pandas, Matplotlib, Seaborn, Math), Jupyter Notebook, Power BI.

4. Exploratory Data Analysis Using Python

Began with data preparation and cleaning in python

- **Data Loading**: Imported the datasets using pandas
- **Initial exploration:** Used df.info() to check structure and .describe() for summary statistics

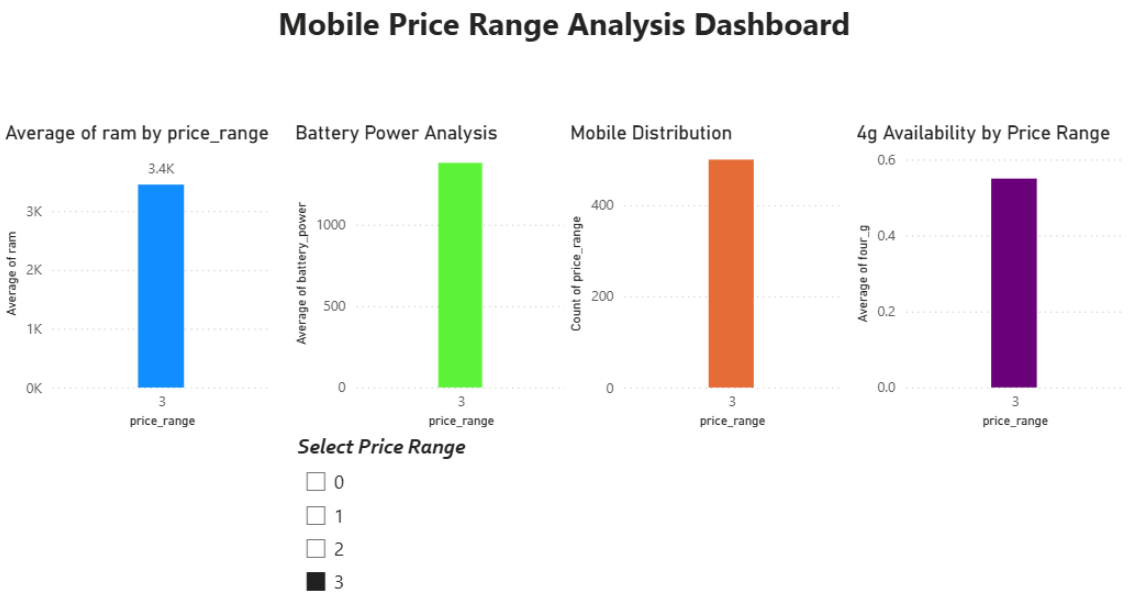
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 21 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   battery_power   2000 non-null   int64
 1   blue            2000 non-null   int64
 2   clock_speed     2000 non-null   float64
 3   dual_sim        2000 non-null   int64
 4   fc              2000 non-null   int64
 5   four_g          2000 non-null   int64
 6   int_memory      2000 non-null   int64
 7   m_dep           2000 non-null   float64
 8   mobile_wt       2000 non-null   int64
 9   n_cores         2000 non-null   int64
 10  pc              2000 non-null   int64
 11  px_height       2000 non-null   int64
 12  px_width        2000 non-null   int64
 13  ram             2000 non-null   int64
 14  sc_h            2000 non-null   int64
 15  sc_w            2000 non-null   int64
 16  talk_time       2000 non-null   int64
 17  three_g         2000 non-null   int64
 18  touch_screen    2000 non-null   int64
 19  wifi            2000 non-null   int64
 20  price_range     2000 non-null   int64
dtypes: float64(2), int64(19)
memory usage: 328.3 KB
```

- Performed distribution analysis and correlation heatmap. Found that RAM has strong correlation with price range

5. Power Bi Dashboard:

Created interactive dashboard with bar charts, average comparisons, and slicers



6. Key Insights

Higher RAM leads to higher price range. Battery power moderately affects pricing. Internal memory also plays a role

7. Business Recommendations

Focus on mid-range segment, improve feature combinations, and target high RAM devices.

8. Limitations

Dataset is limited and does not include real-world factors like brand value or marketing.

9. Conclusion

The project successfully provides insights into mobile pricing and supports better business decisions.